

Assignment 5 CSE 16-02: Applied Discrete Mathematics Winter 2026

Instructor: Cedric Westphal
Due February 12th, 2026 11:59 PM

Complete the questions carefully and show your work. Please remember to:

- Scan the pages and combine them into a single PDF file. Make sure your work is readable and that pages are oriented correctly.
- Upload your PDF file to Canvas/Gradescope under Assignment 5.
- In Gradescope select the page(s) for each problem.

Notation: $\emptyset$ is the empty set, $\mathbb{N}$ is the set of natural numbers, $\mathbb{Z}$ is the set of integers, $\mathbb{Q}$ is the set of rational numbers, and $\mathbb{R}$ is the set of real numbers. $\mathcal{P}(A)$ is the power set of A .

[46 points total in assignment.]

0. Each answer is clearly written and each answer is correctly mapped to its rubric on Gradescope. [4 points]
1. For $a, b \in \mathbb{N}$, prove the following statements below. Hint: to prove that g is the $\gcd(a, b)$ you need to prove that $g|a$ and $g|b$ and that g is the greatest such common divisor. **[10 points]**
 - (a) Let $a = 5x$ and $b = 5y + 3$ for some x and y in $\mathbb{N}$. Prove that $\gcd(a, b) = \gcd(x, b)$. Write a proof that shows: (i) that $\gcd(a, b)|x$, then (ii) show that it implies that $\gcd(a, b)$ equals $\gcd(x, b)$. **[2 points]**
 - (b) If both a and b are divisible by 3, then $\gcd(a, b) = 3 \gcd(a/3, b/3)$. **[2 points]**
 - (c) If m is a common divisor of a and b , show the general case of the previous question, namely that $\gcd(a, b) = m \gcd(a/m, b/m)$. **[2 points]**
 - (d) Prove that $\gcd(a, b) = \gcd(a + b, b)$. **[2 points]**
 - (e) Is it true or false that, for $a > b$, $\gcd(a, b) = \gcd(a + b, a - b)$? Give a clear explanation of your reasoning. **[2 points]**
2. We will prove that if n is an odd integer $\in \mathbb{N}$, then 8 divides $n^6 - 1$. We're going to do this in two ways. **[7 points]**
 - (a) First, prove that if n is an odd integer $\in \mathbb{N}$, then 8 divides $n^2 - 1$.
 - (b) Prove that if n is an odd integer, then so is n^3 .
 - (c) Prove that 8 divides $n^6 - 1$ for n odd, using the previous questions.
 - (d) For the second way to prove this, express $n^6 - 1$ for n odd using the binomial theorem for $n = 2k + 1$ as a sum of terms of the form $a_i k^i$? Write this sum.
 - (e) Consider the last two terms for $a_2 k^2$ and $a_1 k$. Can you rewrite $a_2 k^2 + a_1 k$ as $b k^2 + c(k^2 + k)$ where b is divisible by 8?
 - (f) Prove that 8 divides $c(k^2 + k)$
 - (g) Prove the result stated at the beginning of this Problem, using the previous three steps.
3. Prove or disprove: $\forall x \in \mathbb{R}, \sqrt{x} \notin \mathbb{Q} \rightarrow x \notin \mathbb{Q}$. **[5 points]**
4. Prove or disprove: $\sqrt{3} \in \mathbb{Q}$. **[5 points]**
5. Prove: If $n, m \in \mathbb{Z}$ and $7n^2 + 11m^3 = 325,014$ then n and m have the same parity. **[5 points]**
6. Prove or disprove: If $x \notin \mathbb{Q}$, $y \in \mathbb{Q}$, $x \neq 0$, and $y \neq 0$, then for all $n, m \in \mathbb{N}$, $x^n y^m \notin \mathbb{N}$. **[5 points]**
7. Consider the sets

$$S_1 = \{k^2 + 3 : k \text{ is an even natural number}\}, S_2 = \{4k + 3 : k \in \mathbb{N}\}.$$

- (a) Prove $S_1 \subseteq S_2$. **[3 points]**
- (b) Prove $\neg(S_2 \subseteq S_1)$. **[2 points]**